

2.29 Numerical Fluid Mechanics

Spring 2015 – Lecture 18

REVIEW Lecture 17:

- End of Finite Volume Methods – Cartesian grids

- Higher order (interpolation) schemes

- Solution of the Navier-Stokes Equations

$$\frac{\partial \rho \vec{v}}{\partial t} + \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + \mu \nabla^2 \vec{v} + \rho \vec{g}$$

$$\nabla \cdot \vec{v} = 0$$

- Discretization of the convective and viscous terms

- Discretization of the pressure term

$$\tilde{p} = p - \rho \mathbf{g} \cdot \mathbf{r} + \mu \frac{2}{3} \nabla \cdot \mathbf{u} \quad (p \vec{e}_i - \rho g_i x_i \vec{e}_i + \frac{2}{3} \mu \frac{\partial u_j}{\partial x_j} \vec{e}_i)$$

- Conservation principles

$$\int_S -\tilde{p} \vec{e}_i \cdot \vec{n} dS$$

- Momentum and Mass

- Energy

$$\frac{\partial}{\partial t} \int_{CV} \rho \frac{\|\vec{v}\|^2}{2} dV = - \int_{CS} \rho \frac{\|\vec{v}\|^2}{2} (\vec{v} \cdot \vec{n}) dA - \int_{CS} p \vec{v} \cdot \vec{n} dA + \int_{CS} (\vec{\varepsilon} \cdot \vec{v}) \cdot \vec{n} dA + \int_{CV} (-\vec{\varepsilon} : \nabla \vec{v} + p \nabla \cdot \vec{v} + \rho \vec{g} \cdot \vec{v}) dV$$

- Choice of Variable Arrangement on the Grid

- Collocated and Staggered

- Calculation of the Pressure

TODAY (Lecture 18):

Numerical Methods for the Navier-Stokes Equations

- Solution of the Navier-Stokes Equations
 - Discretization of the convective and viscous terms
 - Discretization of the pressure term
 - Conservation principles
 - Choice of Variable Arrangement on the Grid
 - Calculation of the Pressure
 - Pressure Correction Methods
 - A Simple Explicit Scheme
 - A Simple Implicit Scheme
 - Nonlinear solvers, Linearized solvers and ADI solvers
 - Implicit Pressure Correction Schemes for steady problems
 - Outer and Inner iterations
 - Projection Methods
 - Non-Incremental and Incremental Schemes
 - Fractional Step Methods:
 - Example using Crank-Nicholson

References and Reading Assignments

- Chapter 7 on “Incompressible Navier-Stokes equations” of “J. H. Ferziger and M. Peric, Computational Methods for Fluid Dynamics. Springer, NY, 3rd edition, 2002”
- Chapter 11 on “Incompressible Navier-Stokes Equations” of T. Cebeci, J. P. Shao, F. Kafyeke and E. Laurendeau, *Computational Fluid Dynamics for Engineers*. Springer, 2005.
- Chapter 17 on “Incompressible Viscous Flows” of Fletcher, *Computational Techniques for Fluid Dynamics*. Springer, 2003.

Calculation of the Pressure

The Navier-Stokes equations do not have an independent equation for pressure

- But the pressure gradient contributes to each of the three momentum equations
- For incompressible fluids, mass conservation becomes a kinematic constraint on the velocity field: we then have no dynamic equations for both density and pressure
- For compressible fluids, mass conservation is a dynamic equation for density
 - Pressure can then be computed from density using an equation of state
 - For incompressible flows (or low Mach numbers), density is not a state variable, hence can't be solved for
- For incompressible flows:
 - Momentum equations lead to the velocities $\Rightarrow$
 - Continuity equation should lead to the pressure, but it does not contain pressure! How can p be estimated?

Calculation of the Pressure

- Navier-Stokes, incompressible: $\frac{\partial \rho \vec{v}}{\partial t} + \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + \mu \nabla^2 \vec{v} + \rho \vec{g}$
 $\nabla \cdot \vec{v} = 0$
- Combine the two conservation eqs. to obtain an equation for p
 - Since the cons. of mass has a divergence form, take the divergence of the momentum equation, using cons. of mass:
 - For constant viscosity and density:

$$\nabla \cdot \nabla p = \nabla^2 p = -\nabla \cdot \frac{\partial \rho \vec{v}}{\partial t} - \nabla \cdot (\nabla \cdot (\rho \vec{v} \vec{v})) + \nabla \cdot (\mu \nabla^2 \vec{v}) + \nabla \cdot (\rho \vec{g}) = -\nabla \cdot (\nabla \cdot (\rho \vec{v} \vec{v}))$$

- This pressure equation is elliptic (Poisson eqn. once velocity is known)
 - It can be solved by methods we have seen earlier for elliptic equations

$$\frac{\partial}{\partial x_i} \left(\frac{\partial p}{\partial x_i} \right) = - \frac{\partial}{\partial x_i} \left(\frac{\partial (\rho u_i u_j)}{\partial x_j} \right)$$

– Important Notes

- RHS: Terms inside divergence (derivatives of momentum terms) must be approximated in a form consistent with that of momentum eqns. However, divergence is that of cons. of mass.
- Laplacian operator comes from i) divergence of cons. of mass and ii) gradient in momentum eqns.: consistency must be maintained, i.e. divergence and gradient discrete operators in Laplacian should be those of the cons. of mass and of the momentum eqns., respectively
- Best to derive pressure equation from discretized momentum/continuity equations

Pressure-correction Methods

- First solve the momentum equations to obtain the velocity field for a known pressure
- Then solve the Poisson equation to obtain an updated/corrected pressure field
- Another way: modify the continuity equation so that it becomes hyperbolic (even though it is elliptic)
 - Artificial Compressibility Methods
- Notes:
 - The general pressure-correction method is independent of the discretization chosen for the spatial derivatives $\Rightarrow$ in theory any discretization can be used
 - We keep density in the equations (flows are assumed incompressible, but small density variations are considered)

A Simple Explicit Time Advancing Scheme

- Simple method to illustrate how the numerical Poisson equation for the pressure is constructed and the role it plays in enforcing continuity
- Specifics of spatial derivative scheme not important, hence, we look at the equation discretized in space, but not in time.

- Use $\frac{\delta}{\delta x_i}$ to denote discrete spatial derivatives.

This gives:
$$\frac{\partial \rho u_i}{\partial t} = -\frac{\partial(\rho u_i u_j)}{\partial x_j} - \frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} \Rightarrow \frac{\partial \rho u_i}{\partial t} = -\frac{\delta(\rho u_i u_j)}{\delta x_j} - \frac{\delta p}{\delta x_i} + \frac{\delta \tau_{ij}}{\delta x_j} = H_i - \frac{\delta p}{\delta x_i}$$



Note: $p = p_{real} - \rho g_i x_i$

- Simplest approach: Forward Euler for time integration, which gives:

$$(\rho u_i)^{n+1} - (\rho u_i)^n = \Delta t \left(H_i^n - \frac{\delta p^n}{\delta x_i} \right)$$

- In general, the new velocity field we obtain at time $n+1$ does not satisfy the discrete continuity equation:

$$\frac{\delta(\rho u_i)^{n+1}}{\delta x_i} = 0$$

A Simple Explicit Time Advancing Scheme

- How can we enforce continuity at $n+1$?
- Take the discrete numerical divergence of the NS eqs.:

$$(\rho u_i)^{n+1} - (\rho u_i)^n = \Delta t \left(H_i^n - \frac{\delta p^n}{\delta x_i} \right) \Rightarrow \boxed{\frac{\delta(\rho u_i)^{n+1}}{\delta x_i} - \frac{\delta(\rho u_i)^n}{\delta x_i} = \Delta t \left[\frac{\delta}{\delta x_i} \left(H_i^n - \frac{\delta p^n}{\delta x_i} \right) \right]}$$

- The first term is the divergence of the new velocity field, which we want to be zero, so we set it to zero.
- Second term is zero if continuity was enforced at time step n
- Third term can be zero or not, but the two above conditions set it to zero
- All together, we obtain:
$$\boxed{\frac{\delta}{\delta x_i} \left(\frac{\delta p^n}{\delta x_i} \right) = \frac{\delta H_i^n}{\delta x_i}}$$
- Note that this includes the divergence operator from the continuity eqn. (outside) and the pressure gradient from the momentum equation (inside)
- Pressure gradient could be explicit (n) or implicit ($n+1$)

A Simple Explicit Time Advancing Scheme: Summary of the Algorithm

- Start with velocity at time t_n which is divergence free
- Compute RHS of pressure equation at time t_n
- Solve the Poisson equation for the pressure at time t_n
- Compute the velocity field at the new time step using the momentum equation: It will be *discretely* divergence free
- Continue to next time step

A Simple Implicit Time Advancing Scheme

Additional difficulties arise when an implicit method is used to solve the (incompressible) NS equations

- To illustrate, let's first try the simplest: backward/implicit Euler

– Recall:

$$\frac{\partial \rho u_i}{\partial t} = -\frac{\delta(\rho u_i u_j)}{\delta x_j} - \frac{\delta p}{\delta x_i} + \frac{\delta \tau_{ij}}{\delta x_j} = H_i - \frac{\delta p}{\delta x_i}$$

– Implicit Euler:

$$(\rho u_i)^{n+1} - (\rho u_i)^n = \Delta t \left(H_i^{n+1} - \frac{\delta p^{n+1}}{\delta x_i} \right) = \Delta t \left(-\frac{\delta(\rho u_i u_j)^{n+1}}{\delta x_j} + \frac{\delta \tau_{ij}^{n+1}}{\delta x_j} - \frac{\delta p^{n+1}}{\delta x_i} \right)$$

- Difficulties (specifics for incompressible case)

1) Set numerical divergence of velocity field at new time-step to be zero

- Take divergence of momentum, assume velocity is divergence-free at time t_n and demand zero divergence at t_{n+1} . This leads to:

$$\frac{\delta(\rho u_i)^{n+1}}{\delta x_i} - \frac{\delta(\rho u_i)^n}{\delta x_i} = \Delta t \left[\frac{\delta}{\delta x_i} \left(H_i^{n+1} - \frac{\delta p^{n+1}}{\delta x_i} \right) \right] \Rightarrow \frac{\delta}{\delta x_i} \left(\frac{\delta p^{n+1}}{\delta x_i} \right) = \frac{\delta}{\delta x_i} \left(-\frac{\delta(\rho u_i u_j)^{n+1}}{\delta x_j} + \frac{\delta \tau_{ij}^{n+1}}{\delta x_j} \right)$$

- Problem: The RHS can not be computed until velocities are known at t_{n+1} (and these velocities can not be computed until p^{n+1} is available)
- Result: Poisson and momentum equations have to be solved simultaneously

A Simple Implicit Time Advancing Scheme, Cont'd

2) Even if p^{n+1} known, a large system of nonlinear momentum equations must be solved for the velocity field:

$$(\rho u_i)^{n+1} - (\rho u_i)^n = \Delta t \left(-\frac{\delta(\rho u_i u_j)^{n+1}}{\delta x_j} + \frac{\delta \tau_{ij}^{n+1}}{\delta x_j} - \frac{\delta p^{n+1}}{\delta x_i} \right)$$

Three approaches for solution:

– First approach: nonlinear solvers

- Use velocities at t_n for initial guess of u_i^{n+1} (or use explicit-scheme as first guess) and then employ a nonlinear solver (Fixed-point, Newton-Raphson or Secant methods) at each time step
- Nonlinear solver is applied to the nonlinear algebraic equations

$$(\rho u_i)^{n+1} - (\rho u_i)^n = \Delta t \left(-\frac{\delta(\rho u_i u_j)^{n+1}}{\delta x_j} + \frac{\delta \tau_{ij}^{n+1}}{\delta x_j} - \frac{\delta p^{n+1}}{\delta x_i} \right)$$

$$\frac{\delta}{\delta x_i} \left(\frac{\delta p^{n+1}}{\delta x_i} \right) = \frac{\delta}{\delta x_i} \left(-\frac{\delta(\rho u_i u_j)^{n+1}}{\delta x_j} + \frac{\delta \tau_{ij}^{n+1}}{\delta x_j} \right)$$

A Simple Implicit Time Advancing Scheme, Cont'd

- Second approach: linearize the equations about the result at t_n

$$u_i^{n+1} = u_i^n + \Delta u_i \quad \Rightarrow$$

$$u_i^{n+1} u_j^{n+1} = u_i^n u_j^n + u_i^n \Delta u_j + u_j^n \Delta u_i + \Delta u_i \Delta u_j$$

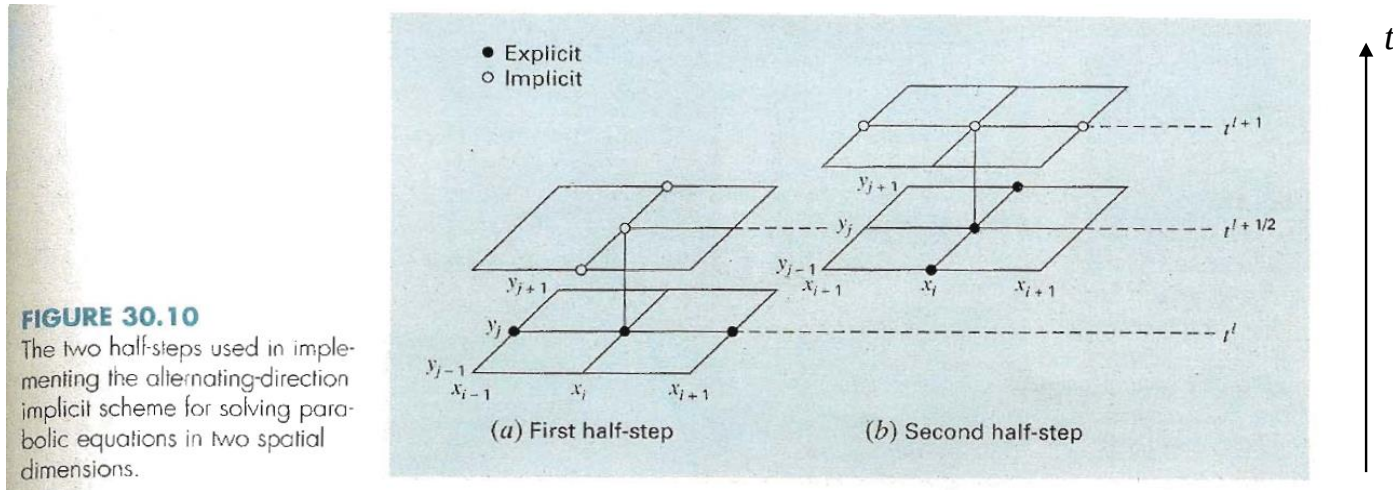
- We'd expect the last term to be of 2nd order in Δt , it can thus be neglected (for a 2nd order in time, e.g. C-N scheme, it would still be of same order as spatial discretization error, so can still be neglected).
- Hence, doing the same in the other terms, the (incompressible) momentum equations are then approximated by:

$$(\rho u_i)^{n+1} - (\rho u_i)^n = \rho \Delta u_i = \Delta t \left(-\frac{\delta(\rho u_i u_j)^n}{\delta x_j} - \frac{\delta(\rho u_i^n \Delta u_j)}{\delta x_j} - \frac{\delta(\rho \Delta u_i u_j^n)}{\delta x_j} + \frac{\delta \tau_{ij}^n}{\delta x_j} + \frac{\delta \Delta \tau_{ij}}{\delta x_j} - \frac{\delta p^n}{\delta x_i} - \frac{\delta \Delta p}{\delta x_i} \right)$$

- One then solves for Δu_i and Δp (using the above mom. eqn. and its Δp eqn.)
 - This linearization takes advantage of the fact that the nonlinear term is only quadratic
 - However, a large coupled linear system (Δu_i & Δp) still needs to be inverted. Direct solution is not recommended: use an iterative scheme
- A third interesting solution scheme: an Alternate Direction Implicit scheme

Parabolic PDEs: Two spatial dimensions (from Lecture 14)

ADI scheme (Two Half steps in time)



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Source: Chapra, S., and R. Canale. *Numerical Methods for Engineers*. McGraw-Hill, 2005.

- 1) From time n to $n+1/2$: Approximation of 2nd order x derivative is explicit, while the y derivative is implicit. Hence, tri-diagonal matrix to be solved:

$$\frac{T_{i,j}^{n+1/2} - T_{i,j}^n}{\Delta t / 2} = c^2 \frac{T_{i-1,j}^n - 2T_{i,j}^n + T_{i+1,j}^n}{\Delta x^2} + c^2 \frac{T_{i,j-1}^{n+1/2} - 2T_{i,j}^{n+1/2} + T_{i,j+1}^{n+1/2}}{\Delta y^2} \quad (O(\Delta x^2 + \Delta y^2))$$

- 2) From time $n+1/2$ to $n+1$: Approximation of 2nd order x derivative is implicit, while the y derivative is explicit. Another tri-diagonal matrix to be solved:

$$\frac{T_{i,j}^{n+1} - T_{i,j}^{n+1/2}}{\Delta t / 2} = c^2 \frac{T_{i-1,j}^{n+1} - 2T_{i,j}^{n+1} + T_{i+1,j}^{n+1}}{\Delta x^2} + c^2 \frac{T_{i,j-1}^{n+1/2} - 2T_{i,j}^{n+1/2} + T_{i,j+1}^{n+1/2}}{\Delta y^2} \quad (O(\Delta x^2 + \Delta y^2))$$

Parabolic PDEs: Two spatial dimensions

ADI scheme (Two Half steps in time)

(from Lecture 14)

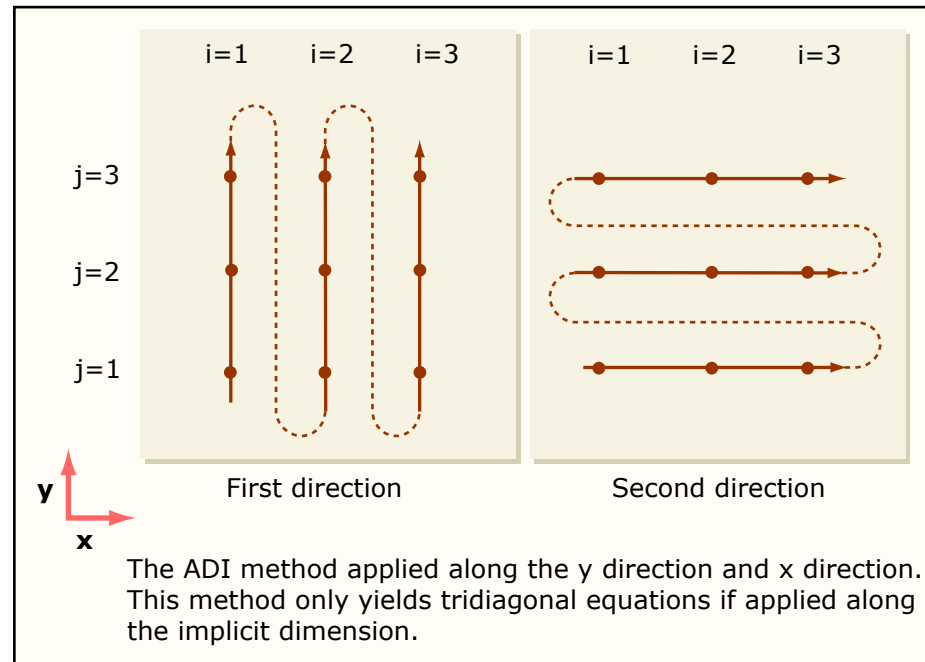


Image by MIT OpenCourseWare. After Chapra, S., and R. Canale. *Numerical Methods for Engineers*. McGraw-Hill, 2005.

For $\Delta x = \Delta y$:

1) From time n to $n+1/2$:
(1st tri-diagonal sys.)

$$-rT_{i,j-1}^{n+1/2} + 2(1+r)T_{i,j}^{n+1/2} - rT_{i,j+1}^{n+1/2} = rT_{i-1,j}^n + 2(1-r)T_{i,j}^n + rT_{i+1,j}^n$$

2) From time $n+1/2$ to $n+1$:
(2nd tri-diagonal sys.)

$$-rT_{i-1,j}^{n+1} + 2(1+r)T_{i,j}^{n+1} - rT_{i+1,j}^{n+1} = rT_{i,j-1}^{n+1/2} + 2(1-r)T_{i,j}^{n+1/2} + rT_{i,j+1}^{n+1/2}$$

A Simple Implicit Time Advancing Scheme, Cont'd

nate Direction Implicit method

- Split the NS momentum equations into a series of 1D problems, e.g. each being block tri-diagonal. Then, either:
- ADI nonlinear: iterate for the nonlinear terms, or,
- ADI with a local linearization:

- Δp can first be set to zero to obtain a new velocity u_i^* which does not satisfy continuity:

$$(\rho u_i^*)^{n+1} - (\rho u_i)^n = \Delta t \left(-\frac{\delta(\rho u_i u_j)^n}{\delta x_j} - \frac{\delta(\rho u_i^n \Delta u_j)}{\delta x_j} - \frac{\delta(\rho \Delta u_i u_j^n)}{\delta x_j} + \frac{\delta \tau_{ij}^n}{\delta x_j} + \frac{\delta \Delta \tau_{ij}}{\delta x_j} - \frac{\delta p^n}{\delta x_i} \right)$$

- Solve a Poisson equation for the pressure correction. Taking the divergence of:

$$(\rho u_i)^{n+1} - (\rho u_i)^n = \Delta t \left(-\frac{\delta(\rho u_i u_j)^n}{\delta x_j} - \frac{\delta(\rho u_i^n \Delta u_j)}{\delta x_j} - \frac{\delta(\rho \Delta u_i u_j^n)}{\delta x_j} + \frac{\delta \tau_{ij}^n}{\delta x_j} + \frac{\delta \Delta \tau_{ij}}{\delta x_j} - \frac{\delta p^n}{\delta x_i} - \frac{\delta \Delta p}{\delta x_i} \right)$$

$$\Leftrightarrow (\rho u_i)^{n+1} = (\rho u_i^*)^{n+1} - \Delta t \frac{\delta \Delta p}{\delta x_i}$$

gives, $\frac{\delta}{\delta x_i} \left(\frac{\delta \Delta p}{\delta x_i} \right) = \frac{1}{\Delta t} \frac{\delta(\rho u_i^*)^{n+1}}{\delta x_i}$, from which Δp can be solved for.

- Finally, update the velocity: $(\rho u_i)^{n+1} = (\rho u_i^*)^{n+1} - \Delta t \frac{\delta \Delta p}{\delta x_i}$

Methods for solving (steady) NS problems:

Implicit Pressure-Correction Methods

- Simple implicit approach based on linearization is most useful for unsteady problems (with limited time-steps)
 - It is not accurate for large (time) steps (because the linearization would then lead to a large error)
 - Thus, it should not be used for steady problems (which often use large time-steps)
- Steady problems are often solved with an implicit method (with pseudo-time), but with large time steps (no need to reproduce the pseudo-time history)
 - The aim is to rapidly converge to the steady nonlinear solution
- Many steady-state solvers are based on variations of the implicit schemes
 - They use a pressure (or pressure-correction) equation to enforce continuity at each “pseudo-time” steps, also called “outer iteration”

Methods for solving (steady) NS problems: Implicit Pressure-Correction Methods, Cont'd

- For a fully implicit scheme, the steady state momentum equations are:

$$(\rho u_i)^{n+1} - (\rho u_i)^n = 0 \Rightarrow \left(-\frac{\delta(\rho u_i u_j)^{n+1}}{\delta x_j} + \frac{\delta \tau_{ij}^{n+1}}{\delta x_j} - \frac{\delta p^{n+1}}{\delta x_i} \right) = 0$$

- With the discretized matrix notation, the result is a nonlinear algebraic system

$$\mathbf{A}^{\mathbf{u}_i^{n+1}} \mathbf{u}_i^{n+1} = \mathbf{b}_{\mathbf{u}_i}^{n+1} - \frac{\delta p^{n+1}}{\delta x_i}$$

- The $\mathbf{b}$ term in the RHS contains all terms that are explicit (in $\mathbf{u}_i^n$) or linear in $\mathbf{u}_i^{n+1}$ or that are coefficients function of other variables at t_{n+1} , e.g. temperature
- Pressure gradient is still written in symbolic matrix difference form to indicate that any spatial derivatives can be used
- The algebraic system is nonlinear. Again, nonlinear iterative solvers can be used. For steady flows, the tolerance of the convergence of these nonlinear-solver iterations does not need to be as strict as for a true time-marching scheme
- Note two types of successive iterations can be employed with pressure-correction:
 - Outer iterations: (over one pseudo-time step) use nonlinear solvers which update the elements of matrix $\mathbf{A}^{\mathbf{u}_i^{n+1}}$ as well as $\mathbf{u}_i^{n+1}$ (uses no or approx. pressure term, then corrects it)
 - Inner Iterations: linear algebra to solve the linearized system with fixed coefficients

Methods for solving (steady) NS problems: Implicit Pressure-Correction Methods, Cont'd

iteration m (pseudo-time): nonlinear solvers which update the elements of the matrix $\mathbf{A}^{\mathbf{u}_i^{m*}}$ as well as $\mathbf{u}_i^{m*}$:

$$\boxed{\mathbf{A}^{\mathbf{u}_i^{m*}} \mathbf{u}_i^{m*} = \mathbf{b}_{\mathbf{u}_i^{m*}}^{m-1} - \frac{\delta p^{m-1}}{\delta x_i}} \Rightarrow \text{formally, } \mathbf{u}_i^{m*} = \underbrace{\left(\mathbf{A}^{\mathbf{u}_i^{m*}}\right)^{-1} \mathbf{b}_{\mathbf{u}_i^{m*}}^{m-1}}_{\text{best estimate of exact } \mathbf{u} \text{ without any p-grad.}} - \left(\mathbf{A}^{\mathbf{u}_i^{m*}}\right)^{-1} \frac{\delta p^{m-1}}{\delta x_i} \equiv \underline{\tilde{\mathbf{u}}_i^{m*}} - \left(\mathbf{A}^{\mathbf{u}_i^{m*}}\right)^{-1} \frac{\delta p^{m-1}}{\delta x_i}$$

- The resulting velocities $\mathbf{u}_i^{m*}$ do not satisfy continuity (hence the $*$) since the RHS is obtained from p^{m-1} at the end of the previous outer iteration $\rightarrow$ needs to correct $\mathbf{u}_i^{m*}$.
- The final $\mathbf{u}_i^m$ needs to satisfy: $\mathbf{A}^{\mathbf{u}_i^m} \mathbf{u}_i^m = \mathbf{b}_{\mathbf{u}_i^m}^m - \frac{\delta p^m}{\delta x_i}$ and $\frac{\delta \mathbf{u}_i^m}{\delta x_i} = 0 \Rightarrow$

$$\begin{aligned} \mathbf{u}_i^m &= \left(\mathbf{A}^{\mathbf{u}_i^m}\right)^{-1} \mathbf{b}_{\mathbf{u}_i^m}^{m-1} - \left(\mathbf{A}^{\mathbf{u}_i^m}\right)^{-1} \frac{\delta p^m}{\delta x_i} \\ &\approx \underbrace{\left(\mathbf{A}^{\mathbf{u}_i^{m*}}\right)^{-1} \mathbf{b}_{\mathbf{u}_i^{m*}}^{m-1}}_{\text{best estimate of exact } \mathbf{u} \text{ without any p-grad.}} - \left(\mathbf{A}^{\mathbf{u}_i^{m*}}\right)^{-1} \frac{\delta p^m}{\delta x_i} \Rightarrow \end{aligned}$$

$$0 \approx \frac{\delta \tilde{\mathbf{u}}_i^{m*}}{\delta x_i} - \frac{\delta}{\delta x_i} \left(\left(\mathbf{A}^{\mathbf{u}_i^{m*}}\right)^{-1} \frac{\delta p^m}{\delta x_i} \right)$$

- Inner iteration: After solving a Poisson equation for the pressure, the final velocity is calculated using the inner iteration (fixed coefficient $\mathbf{A}$)
- Finally, increase m to $m+1$ and iterate (outer, then inner)

$$\boxed{\mathbf{A}^{\mathbf{u}_i^{m*}} \mathbf{u}_i^m = \mathbf{b}_{\mathbf{u}_i^{m*}}^m - \frac{\delta p^m}{\delta x_i}}$$

This scheme is a variation of previous time-marching schemes. Main differences: i) no time-variation terms, and, ii) the terms in RHS can be explicit or implicit in outer iteration.

Methods for solving (steady) NS problems:

Projection Methods

- These schemes that first construct a velocity field that does not satisfy continuity, but then correct it using a pressure gradient are called “projection methods”:
 - The divergence producing part of the velocity is “projected out”
- One of the most common methods of this type are the pressure-correction schemes
 - Substitute $\mathbf{u}_i^m = \mathbf{u}_i^{m*} + \mathbf{u}'$ and $p^m = p^{m-1} + p'$ in the previous equations
 - Variations of these pressure-correction methods include:
 - SIMPLE (Semi-Implicit Method for Pressure-Linked Equations) method:
 - Neglects contributions of $\mathbf{u}'$ in the pressure equation
 - SIMPLEC: approximate $\mathbf{u}'$ in the pressure equation as a function of p' (better)
 - SIMPLER and PISO methods: iterate to obtain $\mathbf{u}'$
 - There are many other variations of these methods: all are based on outer and inner iterations until convergence at $m(n+1)$ is achieved.

Projection Methods: Example Scheme 1

Guermond et al, CM-AME-2006

Non-Incremental (Chorin, 1968):

- No pressure term used in predictor momentum equation
- Correct pressure based on continuity
- Update velocity using corrected pressure in momentum equation

$$(\rho u_i^*)^{n+1} = (\rho u_i)^n + \Delta t \left(-\frac{\delta(\rho u_i u_j)^{n+1}}{\delta x_j} + \frac{\delta \tau_{ij}^{n+1}}{\delta x_j} \right) ; \quad (\rho u_i^*)^{n+1} \Big|_{\partial D} = (\text{bc})$$

$$\left. \begin{aligned} (\rho u_i)^{n+1} &= (\rho u_i^*)^{n+1} - \Delta t \frac{\delta p^{n+1}}{\delta x_i} \\ \frac{\delta(\rho u_i)^{n+1}}{\delta x_i} &= 0 \end{aligned} \right\} \Rightarrow$$

$$\frac{\delta}{\delta x_i} \left(\frac{\delta p^{n+1}}{\delta x_i} \right) = \frac{1}{\Delta t} \frac{\delta}{\delta x_i} \left((\rho u_i^*)^{n+1} \right) ; \quad \frac{\delta p^{n+1}}{\delta n} \Big|_{\partial D} = 0$$

$$(\rho u_i)^{n+1} = (\rho u_i^*)^{n+1} - \Delta t \frac{\delta p^{n+1}}{\delta x_i}$$

Note: advection term can be treated:

- implicitly for u^* at $n+1$ (need to iterate then), or,
- explicitly (evaluated with u at n), as in 2d FV code and many others

Projection Methods: Example Scheme 2

Guermond et al, CM-AME-2006

Incremental (Goda, 1979):

- Old pressure term used in predictor momentum equation
- Correct pressure based on continuity: $p^{n+1} = p^n + p'$
- Update velocity using pressure increment in momentum equation

$$(\rho u_i^*)^{n+1} = (\rho u_i)^n + \Delta t \left(-\frac{\delta(\rho u_i u_j)^{n+1}}{\delta x_j} + \frac{\delta \tau_{ij}^{n+1}}{\delta x_j} - \frac{\delta p^n}{\delta x_i} \right) ; \quad (\rho u_i^*)^{n+1} \Big|_{\partial D} = (\text{bc})$$

$$\left. \begin{aligned} (\rho u_i)^{n+1} &= (\rho u_i^*)^{n+1} - \Delta t \frac{\delta(p^{n+1} - p^n)}{\delta x_i} \\ \frac{\delta(\rho u_i)^{n+1}}{\delta x_i} &= 0 \end{aligned} \right\} \Rightarrow \frac{\delta}{\delta x_i} \left(\frac{\delta(p^{n+1} - p^n)}{\delta x_i} \right) = \frac{1}{\Delta t} \frac{\delta}{\delta x_i} \left((\rho u_i^*)^{n+1} \right) ; \quad \frac{\delta(p^{n+1} - p^n)}{\delta n} \Big|_{\partial D} = 0$$

$$(\rho u_i)^{n+1} = (\rho u_i^*)^{n+1} - \Delta t \frac{\delta(p^{n+1} - p^n)}{\delta x_i}$$

Notes:

- this scheme assumes $\mathbf{u}'=0$ in the pressure equation. It is as the SIMPLE method, but without the iterations
- As for the non-incremental scheme, the advection term can be explicit or implicit

Projection Methods: Example Scheme 3

Guermond et al, CM-AME-2006

Rotational Incremental (Timmermans et al, 1996):

- Old pressure term used in predictor momentum equation
- Correct pressure based on continuity: $p^{n+1} = p^n + p' = p^n + \delta p^{n+1} + f(u')$
- Update velocity using pressure increment in momentum equation

$$(\rho u_i^*)^{n+1} = (\rho u_i)^n + \Delta t \left(-\frac{\delta(\rho u_i u_j)^{n+1}}{\delta x_j} + \frac{\delta \tau_{ij}^{n+1}}{\delta x_j} - \frac{\delta p^n}{\delta x_i} \right) ; \quad (\rho u_i^*)^{n+1} \Big|_{\partial D} = (\text{bc})$$

$$\tau_{ij}^{n+1} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

$$\left. \begin{aligned} (\rho u_i)^{n+1} &= (\rho u_i^*)^{n+1} - \Delta t \frac{\delta(\delta p^{n+1})}{\delta x_i} \\ \frac{\delta(\rho u_i)^{n+1}}{\delta x_i} &= 0 \end{aligned} \right\} \Rightarrow$$

$$\frac{\delta}{\delta x_i} \left(\frac{\delta(\delta p^{n+1})}{\delta x_i} \right) = \frac{1}{\Delta t} \frac{\delta}{\delta x_i} \left((\rho u_i^*)^{n+1} \right); \quad \frac{\delta(\delta p^{n+1})}{\delta n} \Big|_{\partial D} = 0$$

$$\begin{aligned} (\rho u_i)^{n+1} &= (\rho u_i^*)^{n+1} - \Delta t \frac{\delta(\delta p^{n+1})}{\delta x_i} \\ p^{n+1} &= p^n + \delta p^{n+1} - \mu \frac{\delta}{\delta x_i} \left((u_i^*)^{n+1} \right) \end{aligned}$$

Notes:

- this scheme accounts for u' in the pressure eqn.
- It can be made into a SIMPLE-like method, if iterations are added
- Again, the advection term can be explicit or implicit. The rotational correction to the left assumes explicit advection

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2.29 Numerical Fluid Mechanics

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